

ELECTRIC CIRCUITS LAB - EE1200

Lab-1 report

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January 19, 2026

1 Arbitrary Function Generator

1. An arbitrary function generator (AFG) is an equipment used to generate various waveforms of desired frequency. The most common waveforms used in AFG are Sine wave, Square wave, Ramp wave and the Pulse waves.
2. Parameters like frequency, amplitude, phase, and DC offset of the wave can be controlled using AFG's

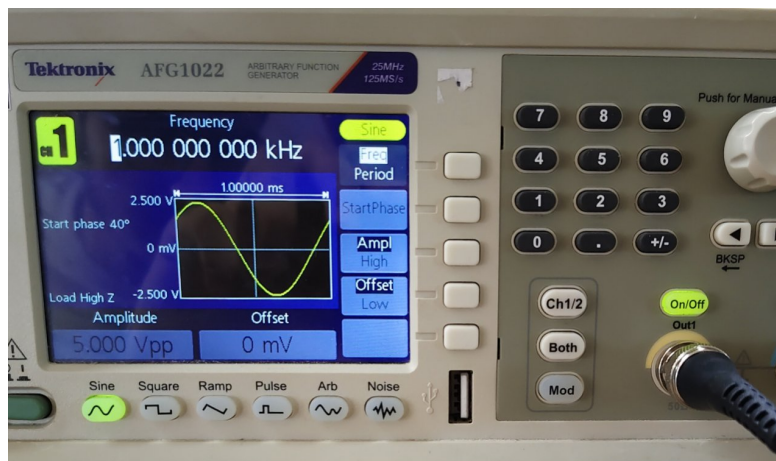


Figure 2: Arbitrary Function Generator with parameters that can be controlled

2 Digital Storage Oscilloscope

A digital storage oscilloscope (DSO) is an oscilloscope which stores and analyzes wave digitally rather than using analog techniques.

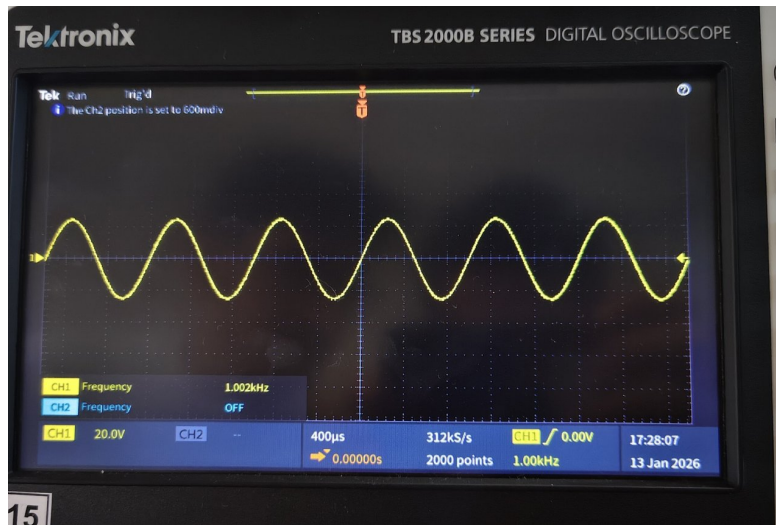


Figure 2: Digital Storage Oscilloscope

2.1 Voltage and time divisions

1. V/div stand for volatge per division sets the scale on the y axis. Time/div stands for time per division sets the scale on the x axis.
2. These are very essential because they control how the waveform's amplitude and duration are scaled to fit the screen. These allow user to measure various parameters like frequency, amplitude, peak to peak ripple, etc.

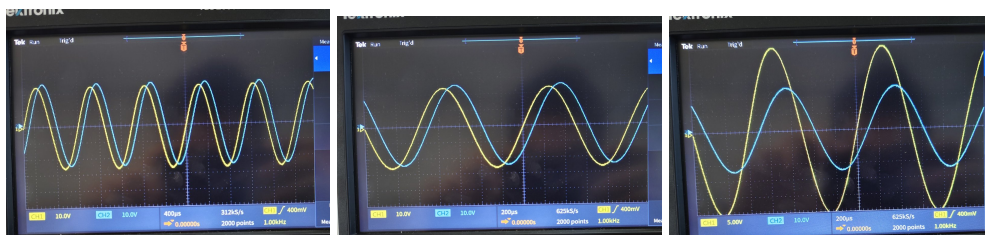


Figure 2: v/div and t/div figures

Explanation:

1. Figure 1 has 10v per division and 400 μ s per division
2. Figure 2, we changed the time per division to 200 μ s. So, the waveform now requires two divisions in the new time scale to cover the distance of one division in the previous scale on x axis. As a result, the number of cycles displayed decreases.
3. In Figure 3, we changed the voltage per division from 10V to 5V. So, the waveform now occupies twice the number of vertical divisions. As a result, the waveform appears vertically expanded.

2.2 XY mode in a DSO

1. In the case of multiple inputs to DSO we use XY mode which plots one input channel voltage vs another input channel voltage.
2. In XY mode X axis is the voltage of channel 1 input and Y axis is the voltage of channel 2 input whereas in time-domain analysis X axis is time and Y axis is input voltage.
3. To enter XY mode in DSO click on Acquire button and click on XY mode.

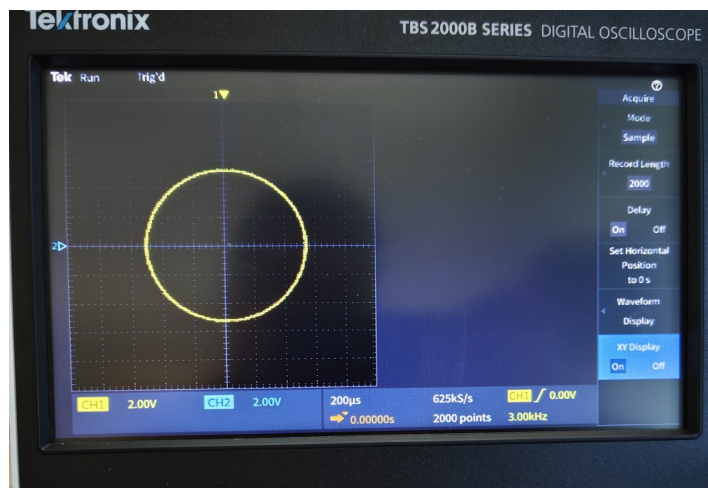


Figure 3: XY mode in DSO

2.3 Capturing single shot events

1. The mode of the function generator is to be shifted to the **Burst Mode** and the trigger is to be set the **External Trigger**.

2. Number of cycles of the waveform are to be appropriately selected.
3. The oscilloscope's coupling should be set DC, and the sweep should be set to **Normal**.



Figure 3: single shot event

2.4 Measuring phase difference between two waves

1. Phase difference is defined as angular displacement between two waves of same frequency.
2. In DSO we have a built-in function to measure phase difference between two waves. Click on measure button on DSO and select phase in it and select ch1-ch2 in it.
3. The DSO will display the phase difference in degrees



Figure 3: Measuring phase difference between two waves

3 Lissajous figures

Lissajous figures are the curves obtained when two periodic signals are applied to the horizontal and vertical inputs of an oscilloscope. These figures depend on the frequency ratio, amplitude ratio, and phase difference of the two signals.

3.1 0° phase difference

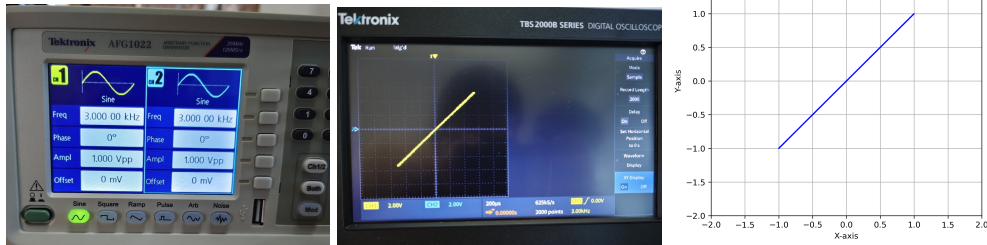


Figure 3: Lissajous Plot with phase 0°

Explanation

$$x = x(t) = \sin t \quad (1)$$

$$y = y(t) = \sin t \quad (2)$$

Locus of (1) and (2) is a straight line through origin

$$x = y \quad (3)$$

The same is also verified by the python plot

3.2 90° phase difference

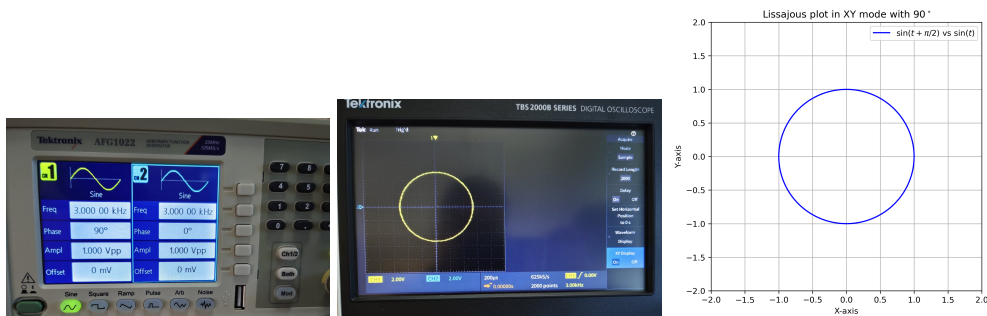


Figure 3: Lissajous Plot with phase 90°

Explanation

$$x = x(t) = \cos t \quad (4)$$

$$y = y(t) = \sin t \quad (5)$$

Locus of (4) and (5) is a circle with centre origin

$$x^2 + y^2 = 1 \quad (6)$$

The same is also verified by the python plot

3.3 To determine frequency ratio

The frequency ratio can be determined by seeing lissajous figures as

$$\text{Ratio} = \frac{\text{No. of loops along vertical axis}}{\text{No. of loops along horizontal axis}} = \frac{N_x}{N_y} \quad (7)$$

Using (7) we can measure the unknown frequency of the wave if we know the frequency of the other wave.

Example 1:



Figure 3: Lissajous Plot to determine frequency ratio

From (7),

$$\text{ratio} = \frac{5}{4} \quad (8)$$

The curve and the ratio are also verified by the above python plot.

Example 2:

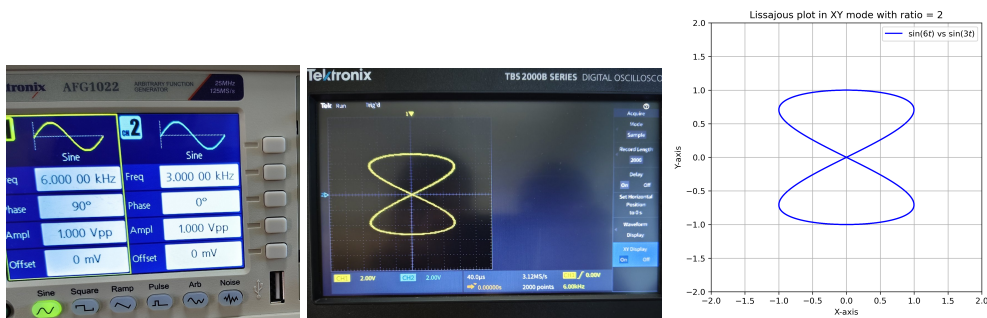


Figure 3: Lissajous Plot to determine frequency ratio

From (7),

$$\text{ratio} = 2 = \frac{6}{3} \quad (9)$$

The curve and the ratio are also verified by the above python plot.

3.4 Practical applications of Lissajous figures

In practice lissajous figures are used in wide variety of applications.

1. To determine the unknown signal frequency and to measure phase difference between the signals
2. To visualize mechanical vibrations and understand how different parts of a machine move relative to each other
3. To observe resonance in coupled oscillators
4. Used in calibrating oscilloscopes and in audio engineering to check stereo balance, signal symmetry, etc.

4 Different Triggering modes in DSO

1. Triggering is the process of aligning the horizontal sweep to a specific point on the input signal to create a stable, usable waveform display
2. There are different modes in DSO such as edge, pulsewidth, runt, etc.
3. **Edge triggering** : It is the most common trigger used. It triggers when the signal crosses a specified voltage in a specific direction—either positive slope or negative slope.

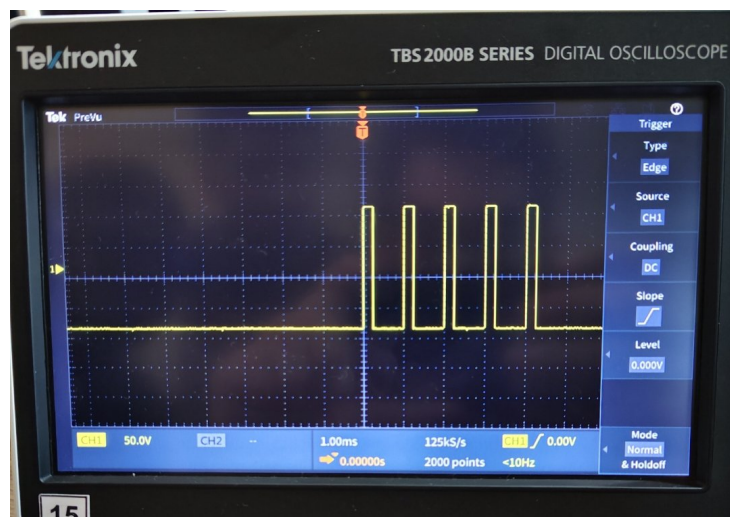


Figure 3: Edge Trigger

4. **Pulse Width Trigger** : It is a trigger based on the duration of a pulse. We can set conditions to capture pulses that are greater than, less than, or equal to a specified time period.

5. Above time period refers to time which pulse is active i.e

$$\text{time period} = (\% \text{ of duty cycle}) * \frac{1}{\text{frequency of the wave}} \quad (10)$$

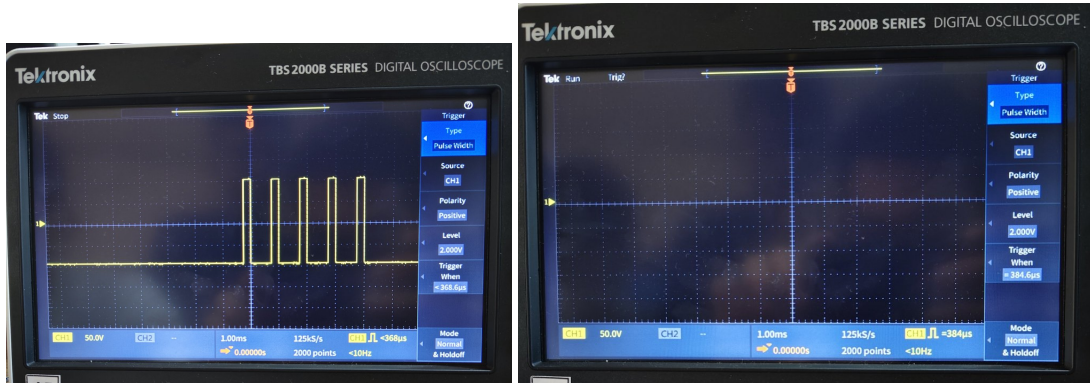


Figure 5: Pulse Width Trigger with 25% duty cycle and 1000Hz frequency

Explanation : From (10),

$$\text{Time period} = \frac{1}{4} * 1000\mu s \quad (11)$$

$$\Rightarrow \text{Time period} = 250\mu s \quad (12)$$

In the 1st wave condition is time period < 368.6µs. According to (12) it is satisfied hence we can see the pulse in DSO.

In the 2nd wave condition is time period = 368.6µs. According to (12) it is not satisfied hence we can't see the pulse in DSO.

6. **Runt Trigger :** Captures pulses that cross one voltage threshold but fail to cross a second one before returning.

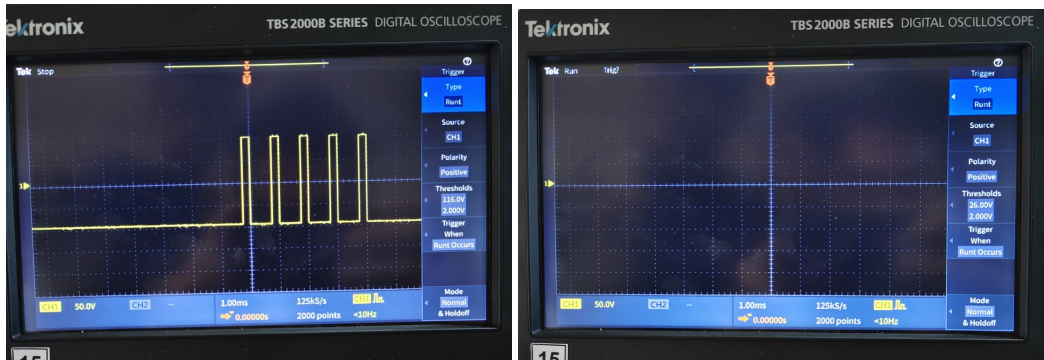


Figure 6: Runt Trigger with different conditions

7. **Level Triggering** :Activates the sweep based on the signal being at a specific logic state (HIGH or LOW)

5 Plots :

These plots follow the order AFG,DSO,Python plot for verification

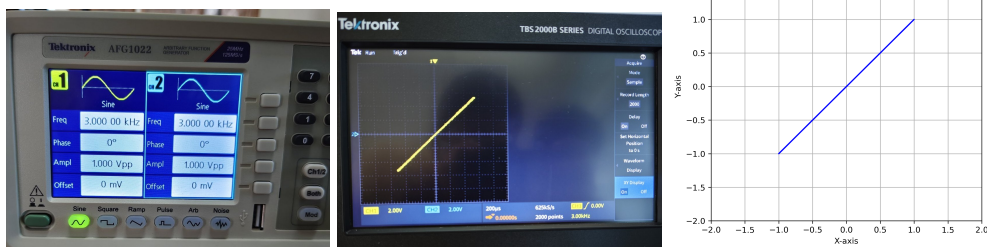


Figure 7: Lissajous Plot1

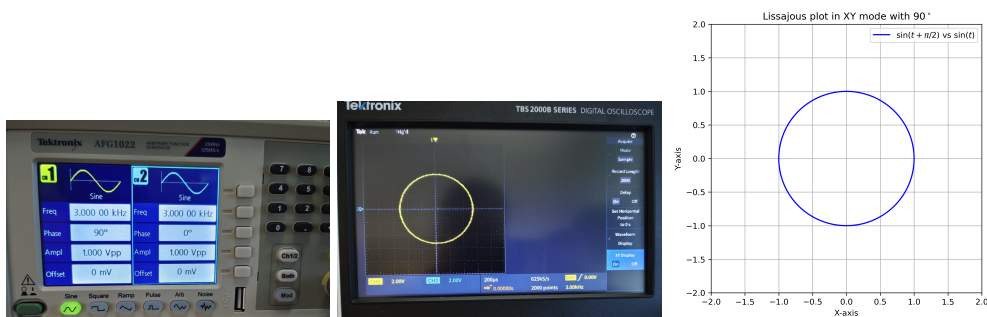


Figure 7: Lissajous Plot2

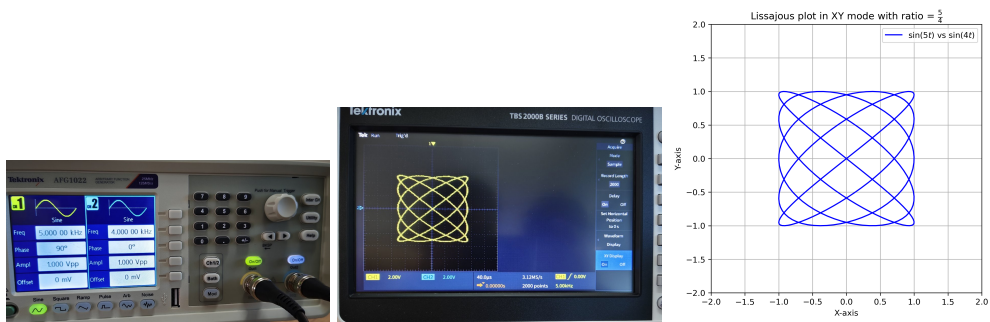


Figure 7: Lissajous Plot3

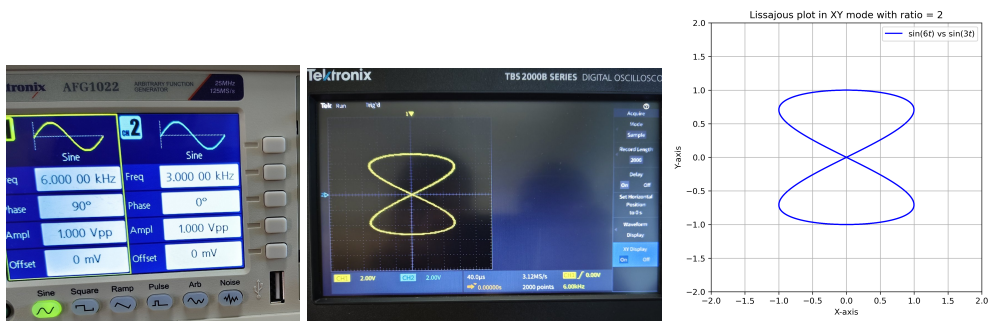


Figure 7: Lissajous Plot4



Figure 7: Lissajous Plot5

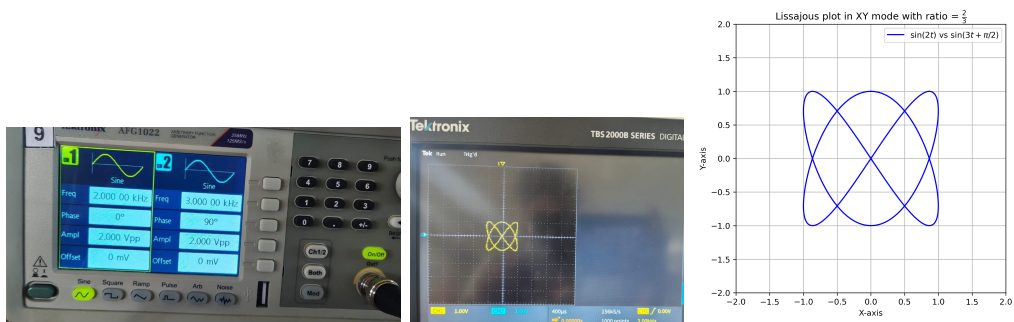


Figure 7: Lissajous Plot6

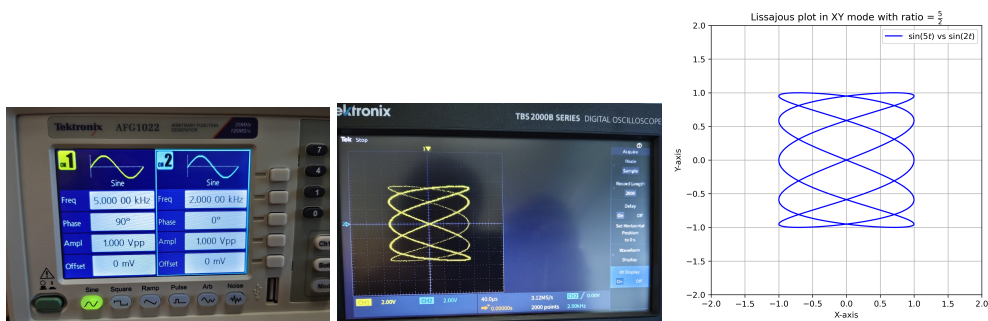


Figure 7: Lissajous Plot7

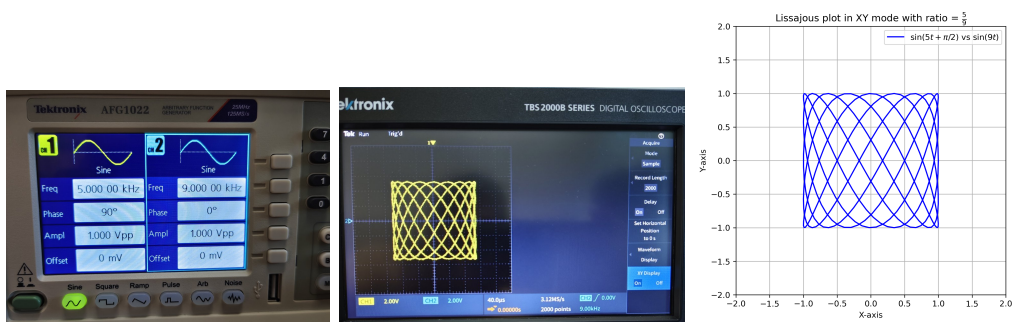


Figure 7: Lissajous Plot8

6 Python code for plots

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 t = np.linspace(0, 2*np.pi, 300)
5
6 fig, ax = plt.subplots(figsize=(6, 6))
7 ax.plot(np.sin(t), np.sin(t), c='blue', label=r'$\sin(t)$ vs $\sin(t)$')
8 ax.set_xlim(-2,2); ax.set_ylim(-2,2)
9 ax.set_xlabel("X-axis"); ax.set_ylabel("Y-axis")
10 ax.set_title(r"Lissajous plot in XY mode with $0^\circ$")
11 ax.legend(); ax.grid(True)
12 plt.savefig("phase_0.png", dpi=300, bbox_inches="tight")
13 plt.clf()
14
15 fig, ax = plt.subplots(figsize=(6, 6))
16 ax.plot(np.cos(t), np.sin(t), c='blue', label=r'$\sin(t + \pi/2)$ vs $\sin(t)$')
17 ax.set_xlim(-2,2); ax.set_ylim(-2,2)
18 ax.set_xlabel("X-axis"); ax.set_ylabel("Y-axis")
19 ax.set_title(r"Lissajous plot in XY mode with $90^\circ$")
20 ax.legend(); ax.grid(True)
21 plt.savefig("phase_90.png", dpi=300, bbox_inches="tight")
22 plt.clf()
23
24 fig, ax = plt.subplots(figsize=(6, 6))
25 ax.plot(np.sin(6*t), np.sin(3*t), c='blue', label=r'$\sin(6t)$ vs $\sin(3t)$')
26 ax.set_xlim(-2,2); ax.set_ylim(-2,2)
27 ax.set_xlabel("X-axis"); ax.set_ylabel("Y-axis")
28 ax.set_title(r"Lissajous plot in XY mode with ratio = 2")
29 ax.legend(); ax.grid(True)
30 plt.savefig("ratio_2.png", dpi=300, bbox_inches="tight")
31 plt.clf()
32
33 fig, ax = plt.subplots(figsize=(6, 6))
34 ax.plot(np.sin(5*t), np.sin(4*t), c='blue', label=r'$\sin(5t)$ vs $\sin(4t)$')
35 ax.set_xlim(-2,2); ax.set_ylim(-2,2)
36 ax.set_xlabel("X-axis"); ax.set_ylabel("Y-axis")
37 ax.set_title(r"Lissajous plot in XY mode with ratio = $\frac{5}{4}$")
38 ax.legend(); ax.grid(True)
39 plt.savefig("ratio_1.png", dpi=300, bbox_inches="tight")
```

```

40 plt.clf()
41
42 fig, ax = plt.subplots(figsize=(6, 6))
43 ax.plot(np.cos(5*t), np.sin(9*t), c='blue', label=r'$\sin(5t + \pi/2)$ vs $\sin(9t)$')
44 ax.set_xlim(-2,2); ax.set_ylim(-2,2)
45 ax.set_xlabel("X-axis"); ax.set_ylabel("Y-axis")
46 ax.set_title(r"Lissajous plot in XY mode with ratio = $\frac{5}{9}$")
47 ax.legend(); ax.grid(True)
48 plt.savefig("plot3.png", dpi=300, bbox_inches="tight")
49 plt.clf()
50
51 fig, ax = plt.subplots(figsize=(6, 6))
52 ax.plot(np.sin(5*t), np.sin(2*t), c='blue', label=r'$\sin(5t)$ vs $\sin(2t)$')
53 ax.set_xlim(-2,2); ax.set_ylim(-2,2)
54 ax.set_xlabel("X-axis"); ax.set_ylabel("Y-axis")
55 ax.set_title(r"Lissajous plot in XY mode with ratio = $\frac{5}{2}$")
56 ax.legend(); ax.grid(True)
57 plt.savefig("plot4.png", dpi=300, bbox_inches="tight")
58 plt.clf()
59
60 fig, ax = plt.subplots(figsize=(6, 6))
61 ax.plot(np.sin(2*t), np.cos(3*t), c='blue', label=r'$\sin(2t)$ vs $\sin(3t + \pi/2)$')
62 ax.set_xlim(-2,2); ax.set_ylim(-2,2)
63 ax.set_xlabel("X-axis"); ax.set_ylabel("Y-axis")
64 ax.set_title(r"Lissajous plot in XY mode with ratio = $\frac{2}{3}$")
65 ax.legend(); ax.grid(True)
66 plt.savefig("plot5.png", dpi=300, bbox_inches="tight")
67 plt.clf()
68
69 fig, ax = plt.subplots(figsize=(6, 6))
70 ax.plot(5*np.sin(t), 2*np.sin(4*t + 5*np.pi/6), c='blue',
71        label=r'$5\sin(t)$ vs $2\sin(4t + 5\pi/6)$')
72 ax.set_xlim(-6,6); ax.set_ylim(-3,3)
73 ax.set_xlabel("X-axis"); ax.set_ylabel("Y-axis")
74 ax.set_title(r"Lissajous plot in XY mode with ratio = $\frac{1}{4}$")
75 ax.legend(); ax.grid(True)
76 plt.savefig("plot6.png", dpi=300, bbox_inches="tight")
77 plt.clf()

```